Functional Equations Solutions

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§1 Walkthrough Problems

Problem 1.1 (USAMO 2002). Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 - y^2) = xf(x) - yf(y)$$

for all pairs of real numbers x and y.

Proof. Let $P(x, y)$ denote the given assertion. Note that $P(0, 0)$ gives us $f(0) = 0$. Then,

$$\begin{gathered} P(x, -x) \implies f(x^2 - x^2) = xf(x) + xf(-x) \\ 0 = x(f(x) + f(-x)) \end{gathered}$$

For nonzero x, we have $f(-x) = -f(x)$, but when $x = 0$ we have $f(0) = 0$ anyway, so we have f is odd for all real numbers x.

$$P(x, 0) \implies f(x^2) = xf(x)$$

So now we can rewrite the original condition as:

$$f(x^2 - y^2) = f(x^2) - f(y^2)$$

We let $a = x^2$, and $b = y^2$ such that, for nonnegative a, b we have:

$$f(a - b) = f(a) - f(b)$$

Again, we can set $a = x_0 + y_0$ and $b = y_0$ to get:

$$f(x_0 + y_0 - y_0) = f(x_0 + y_0) - f(y_0) \implies f(x_0 + y_0) = f(x_0) + f(y_0)$$

for nonnegative x_0, y_0. However, since f is odd, we get that f is additive on the negatives too.

Finally, we let $x = a + b$ in one of the previous equations $f(x^2) = xf(x)$ to get:

$$f(a^2 + 2ab + b^2) = (a + b)(f(a) + f(b))$$

$$f(2ab) = af(b) + bf(a)$$

$$a = 1 \implies f(2b) = f(b) + f(1)b = 2f(b)$$

Which gives $f(x) = cx$ for all x, which upon checking works. □

Problem 1.2 (IMO 2017)**.** Solve over $\mathbb{R}$ the functional equation

$$f(f(x)f(y)) + f(x+y) = f(xy).$$

Proof. We claim that $f(x) = 0$, $f(x) = 1 - x$, and $f(x) = x - 1$ are the only solutions. It's easy to check that these solutions indeed work.

Let $P(x, y)$ denote the given assertion. First, note:

$$P(0,0) \Longrightarrow f(f(0)^2 = 0) \implies \text{ there exists a } z \text{ such that } f(z) = 0.$$

Also, $P(x, 0)$ gives that $f(0) = 0 \implies f \equiv 0$. So from now assume that $f(0) \neq 0$. $\square$